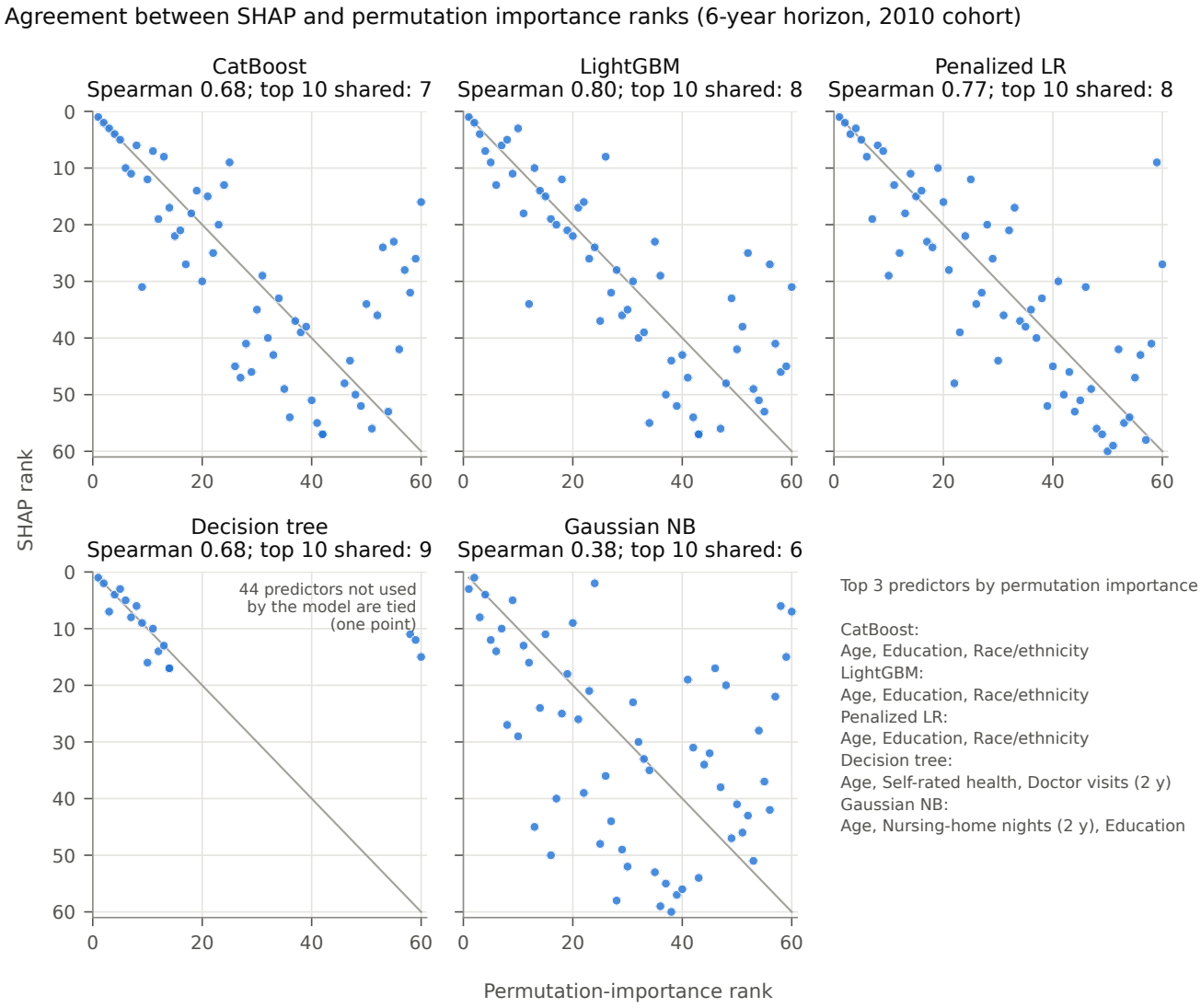
Agreement between SHAP and permutation importance ranks (6-year horizon, 2010 cohort)
CatBoost
Spearman 0.68; top 10 shared: 7
LightGBM
Spearman 0.80; top 10 shared: 8
Penalized LR
Spearman 0.77; top 10 shared: 8
Decision tree
Spearman 0.68; top 10 shared: 9
44 predictors not used by the model are tied (one point)
Gaussian NB
Spearman 0.38; top 10 shared: 6
SHAP rank
Permutation-importance rank
0
10
20
30
40
50
60
Top 3 predictors by permutation importance
CatBoost:
Age, Education, Race/ethnicity
LightGBM:
Age, Education, Race/ethnicity
Penalized LR:
Age, Education, Race/ethnicity
Decision tree:
Age, Self-rated health, Doctor visits (2 y)
Gaussian NB:
Age, Nursing-home nights (2 y), Education